they fuse and become a heavier kind of atom. For all the lightest elements, fusing them together produces radiation, and that radiation is the pressure that holds the star up against collapse.

For a star like the Sun, the outward pressure is provided by the fusion of hydrogen into helium. Most stars are, in fact, just giant helium factories, taking the abundant hydrogen in the universe and sticking it together, countless billions of times per second.

Let's consider the Sun, in particular, for sentimental reasons.

Right now, the Sun is happily burning hydrogen, creating a surplus of helium in its core and causing the temperature and pressure to change over time as the hydrogen-helium balance tips. Because the efficiency of the factory depends on both temperature and pressure, the energy output and size of the Sun will change over time—most noticeably, the Sun will get a little bit more radiant and a little bit bigger[fn10] over the next few million years.

At somewhere around a billion years, we get to the part where we're all fried. But even after the Earth is well on its way to becoming a charred lifeless rock, the Sun has a long way to go yet. As that increased heat is incinerating the inner planets (Mercury and Venus) and evaporating all the oceans off the Earth, so much hydrogen will be burned off that there will only be a shell of hydrogen burning around the central helium-filled core. The core then gets hot enough to start fusing helium into oxygen and carbon and turns the Sun into a huge bloated red giant star. When the Sun eventually runs out of all hydrogen to fuse, a few billion years into its red giant phase, it'll begin its death throes in earnest. The core will start filling with oxygen, then carbon, the production fueled by the squeezing of the core by the gravity of the rest of the star. In the end, though, after the Sun has swelled to engulf the orbit of Venus and the Earth is a smoking ruin, the Sun's gravity won't be enough to maintain the temperatures needed for any further fusion. The outer atmosphere of the star will slough off, and the core will begin to contract.

You might think this would be the end of the Sun—depleted, transformed, and planet-devouring, left with no fusion reactions available that are strong enough to hold it up. Fortunately, there's a kind of pressure even stronger than fusion reactions that can keep the post-red-giant Sun and other stars like it from collapsing entirely, allowing it instead to live out its